MIAE 215 Quiz #1

Please write the following programs by hand using a pen / pencil and paper. Don't use a compiler / editor -- you should compose the program on the paper. After you solve the quiz you should scan in your solutions (using a scanner) or take pictures of them (using a cell phone / camera) and submit them on Moodle.

You have from 5:45 pm to 6:50 pm (65 minutes) to complete the quiz and another 10 minutes to scan and submit the quiz.

All submissions are due at 7 pm (except ACSD students). Late quizzes will not be accepted.

If you are not sure about a question, please state your assumptions / interpretation and proceed to solve the problem based on your assumptions.

1) Write a program that inputs 3 individual characters from the keyboard, one character at a time. If an input is equal to 'x' the program prints out "input is equal to x". The program then prints out each character in the reverse order it was input. For example,

Input from the keyboard (one character at a time):
a (return)
b (return)
c (return)

Output (to the screen):
c
b
a

2) Write a program that calculates y = cos(x) using a for loop for x = 0.0, 1.0e-3, 2.0e-3, …, 1000. It also counts the number of times that y is greater than 0.9 and prints out the final count to the screen.

3) Write a program that estimates the largest possible value for a double by repeatedly multiplying 1.0 by 10 until the number is too large. It does this in two different ways:

i) Printing out the number sequence to the screen (letting a knowledgeable user observe when it gets too large).

ii) Using the fact that a double no longer changes when multiplied by 10 if it becomes too large. At that point print out the message "double out of range" to the screen. Hint: if something doesn't change then it _____ ___ ____.